

[DC]From Being There to Acting There: A Motor Learning Perspective on Reconceptualizing VR Interaction

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ABSTRACT

Virtual Reality promises embodied superpowers that defy physical laws, creating a sensorimotor gap between fixed biological limits and unbounded virtual capabilities. While the field employs various evaluation methods, my systematic review critiques the application of subjective concepts like presence, revealing that its usage is often theoretically underspecified for validating interaction quality. Addressing this, I shift focus from the subjective feeling of *being there* to the computational mechanisms of *acting there*. I conceptualize VR interaction as a sensorimotor mapping and propose a design space grounded in motor learning theory, distinguishing between motor adaptation (recalibrating internal models) and de novo learning (constructing novel policies). Finally, I employ Bayesian computational modeling to decode the learning dynamics of mappings in the design space. By simulating users as Bayesian agents, this work aims to explain the acquisition of virtual capabilities, contributing to a transition from heuristic trial-and-error to predictive computational modeling.

Index Terms: Virtual Reality, interaction techniques, theory, motor learning.

1 INTRODUCTION

The promise of Virtual Reality (VR) lies in interaction mappings that transcend physical laws, granting users capabilities with no real-world antecedent. However, while virtual potential is effectively infinite, it remains bound by the biological bottleneck of the human brain. We define this discrepancy as the sensorimotor gap: the tension between fixed physiological limits and unbounded virtual capabilities.

Currently, the field lacks a framework to explain how users navigate this discrepancy. Existing research has attempted to categorize these techniques through various taxonomies, distinguishing between egocentric vs. exocentric control [2, 5] or mundane vs. magical interactions [7]. Others have proposed frameworks analyzing how techniques alter sensorimotor control loops, separating reality-based interactions from those that are “beyond-real” [1]. However, while these catalogs effectively identify what techniques exist and how they deviate from reality, they rarely explain the underlying learnability of these mappings. They largely remain descriptive, lacking a mechanism to explain why some mappings feel intuitive while others require extensive training.

Evaluation methods face a similar explanatory gap. Traditional metrics like task completion time effectively capture general performance outcomes, yet often fail to reveal the nuances of the adaptation process itself. Specifically, these aggregate measures can miss critical learning behaviors, such as the active refinement of movement required to accommodate new sensorimotor constraints [10]. Subjective evaluation faces similar hurdles. My systematic review [9] reveals that the dominant construct of presence is fre-

quently treated as a theoretically underspecified “magic number” rather than a causal mechanism. Crucially, these static, post-hoc scores fail to capture the dynamic temporal process of learning and adaptation, defining how users master new VR interaction techniques.

Bridging this explanatory gap requires shifting focus from the subjective feeling of being there to the computational mechanisms of acting there. I argue for centering this inquiry on movement mapping—conceptualizing VR interaction as the translation of a finite human control repertoire into an effectively unbounded set of virtual actions. To support this view, I turn to *motor learning theory*, which provides a comprehensive account of how the nervous system acquires and adjusts these mappings [4]. Unlike static outcome metrics, this framework offers trial-level diagnostics—such as aftereffects and generalization—that clarify both the possibilities and biological boundaries of mapping design [3]. Guided by this theoretical perspective, this doctoral work is driven by two fundamental research questions:

Theoretical Framework: How can we structure a causal design space that defines VR interactions by their underlying motor learning processes?

Computational Mechanism: Can we decode the underlying learning process to computationally predict the costs associated with these novel VR interactions?

2 RESEARCH APPROACH: FROM DESIGN SPACE TO COMPUTATIONAL MODELING

To address these questions, this research pivots to the fundamental basis of VR interaction: the sensorimotor mapping. I formalize VR interaction not merely as device usage, but as a mathematical function ($M: X \rightarrow Y$) that routes bodily signals (X) to virtual responses (Y). This formalization allows us to systematically analyze interaction techniques through the lens of motor learning, replacing heuristic trial-and-error with grounded scientific principles.

2.1 A Motor Learning based Design Space

To answer the first research question, we construct a design space starting from the physical body, using motor learning theory to predict the learnability of interaction techniques. This framework distinguishes between mappings that allow for rapid recalibration (adaptation) and those requiring the construction of entirely new control policies (de novo learning) [4, 10]. We identify three primary families of mappings based on how the physical effector is altered:

Homologous Transformations (Φ_h) This family maintains the correspondence between the physical and virtual effector (e.g., hand-to-hand) but alters the transfer function. It covers affine transformations such as gains, rotations, and mirror reversals. The learning mechanism here is ambiguous and spectrum-based: small parametric changes (e.g., small gains) typically trigger adaptation, whereas structural changes (e.g., non-linear rotational gain, mirror reversal) force de novo learning. The boundary is often determined by the magnitude and nature of the distortion.

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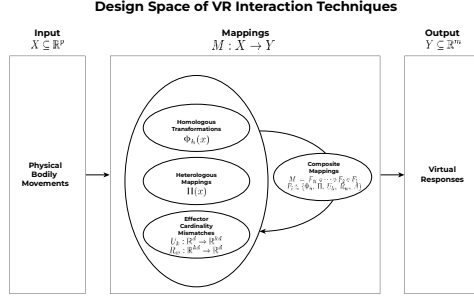


Figure 1: The proposed design space maps physical inputs to virtual outputs through three families, which combine to form composite interaction techniques.

Heterologous Mappings (II) These mappings drive a virtual part using a different physical body part (e.g., foot-to-hand). Because they lack natural anatomical or functional correspondence, they fall strictly into the category of de novo learning tasks. Users cannot adapt an existing model and must instead acquire entirely new coordination patterns to bridge the novel relationship between input and output.

Effector Cardinality Mismatches This family describes mismatches in the number of effectors, such as one-to-many ($U_k: \mathbb{R}^d \rightarrow \mathbb{R}^{kd}$) or many-to-one ($R_w: \mathbb{R}^{kd} \rightarrow \mathbb{R}^d$) mappings. Examples include controlling multiple virtual hands with a single physical hand or two hands jointly control one virtual end-effector. These tasks align with de novo learning by requiring users to reorganize control strategies and perform high-level cognitive remapping to select and engage effectors under new rules.

In this framework, any VR interaction technique can be formalized as a design point—a concrete, ordered series of mapping operators ($M = F_K \circ \dots \circ F_1$). Figure 1 summarizes the space as compositions of family operators. This compositional view reveals that many canonical techniques are effectively blends of operators from distinct families. For example, Go-Go technique [8] can be viewed as combining non-linear arm extension (Φ_h) with effector reassignment (II) to handle reaches beyond anatomical limits. Critically, these composites do not just mix mathematical operators; they interleave distinct learning processes. A single technique may combine components learned via adaptation (e.g., simple gain adjustments) with those requiring de novo learning (e.g., effector reassignment), creating complex learning dynamics that cannot be predicted by observing the components in isolation.

2.2 Decoding Behavior with Bayesian Modeling

To address the second research question, my work moves beyond descriptive analysis to computational explanation, using simulation to rigorously probe the proposed design space. Real-world VR techniques are often composite mappings—intricate blends of operators that simultaneously trigger rapid adaptation and demanding de novo learning. Disentangling these mixed cognitive costs via behavioral observation alone is difficult. Therefore, I employ Bayesian modeling [6], treating the user as an observer constantly minimizing prediction error between internal priors and sensory likelihoods. By instantiating these assumptions into simulated agents, we can decode the learning dynamics of composite mappings, isolating how specific combinations of operators facilitate or impede the acquisition of embodied superpowers. This ultimately

shifts VR design from heuristic trial-and-error to predictive modeling.

3 CONCLUSION

This doctoral research employs a methodological breadth spanning systematic review, theoretical synthesis, and computational modeling to reconceptualize VR interaction. First, my *Systematic Review* challenges the field's reliance on subjective measures, establishing that presence is not a reliable interaction quality metric without a strong underlying theory. Second, I propose a *Motor Learning Based Design Space* that classifies interaction techniques not by metaphor, but by their underlying cognitive mechanisms—specifically distinguishing between adaptation and de novo learning. Finally, I advance *Theoretical Modeling* by using Bayesian models as a promising framework to simulate user behavior and to decode learning process of complex composite mappings. Together, these pillars facilitate a continuous exploration of how VR interaction is conceptualized and studied, contributing to the transition from descriptive phenomenology toward predictive computational modeling.

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